Report_DEB

1. Importation des données

```
file_path <- here::here("mod/FINAL/A_DEB_FLYw")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3303, C2 = 3666, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,          0.1,      0.2,  0.0001,     0.8);
Distrib (pAm,          TruncNormal_cv,       100,      0.2,      40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,       0.8,      0.1,     0.1,       2);
Distrib (v,            TruncNormal_cv,   0.18505,      0.1,   0.001,       2);
Distrib (kap,          Uniform,                0,        1);
Distrib (Ehp,          TruncNormal_cv,       100,      0.2,      20,     300);
Distrib (E_coc,        TruncNormal_cv,        30,      0.2,       1,     300);
Distrib (rOM_ClxHorse, LogUniform,       0.00001,      0.3);
Distrib (dv,           TruncNormal_cv,         2,      0.5,       0,      10);
Distrib (wE,           LogUniform,             1,      1000);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

# f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, Nbl

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                 Fm.1.     pAm.1. r_pAm_pM.1.       v.1.    kap.1.     Ehp.1.
Result_mode 0.12265100 250.319000  1.08695000 0.18498200 0.6442840  23.485100
2.5%        0.12012400 240.761000  0.99119500 0.16048800 0.5241300  21.110200
```

```
97.5%       0.29106985 276.843000  1.13286000 0.22873600 0.7182390  28.917800
Min         0.00117712  61.525200  0.71674700 0.12677300 0.1149050  20.000000
Max         0.29999200 288.509000  1.17480000 0.25747600 0.7685900 130.401000
Result_mean 0.18936840 257.661940  1.06634762 0.19443684 0.6220042  24.469300
Result_sd   0.05047436   9.414489  0.03720417 0.01762878 0.0520435   2.135417
              E_coc.1. rOM_ClxHorse.1.       dv.1.      wE.1. Sigma_W.1.
Result_mode  1.0005400     0.014664900 2.11305000 443.85200 0.16909700
2.5%         1.0016600     0.005813816 0.35896145 280.85000 0.14428900
97.5%        1.2304500     0.015700300 3.98228000 716.83000 0.18579600
Min          1.0000100     0.000142923 0.00018893   2.07827 0.12873500
Max         42.6056000     0.034589500 5.89482000 978.55500 4.86638000
Result_mean  1.0707653     0.010234352 2.10222340 477.40177 0.16394853
Result_sd    0.3145115     0.002840419 0.93949112 113.41856 0.04525606

Number of observations, n =  277
Number of parameters, k =  10
Log-likelihood, LL =  -68.93756
BIC =  194.1153


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.00       1.01
E_coc.1.              1.03       1.03
Ehp.1.                1.03       1.06
Fm.1.                 1.01       1.05
kap.1.                1.25       1.70
pAm.1.                1.03       1.09
r_pAm_pM.1.           1.10       1.31
rOM_ClxHorse.1.       1.02       1.08
Sigma_W.1.            1.01       1.05
v.1.                  1.00       1.01
wE.1.                 1.13       1.40

Multivariate psrf

1.25
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
```

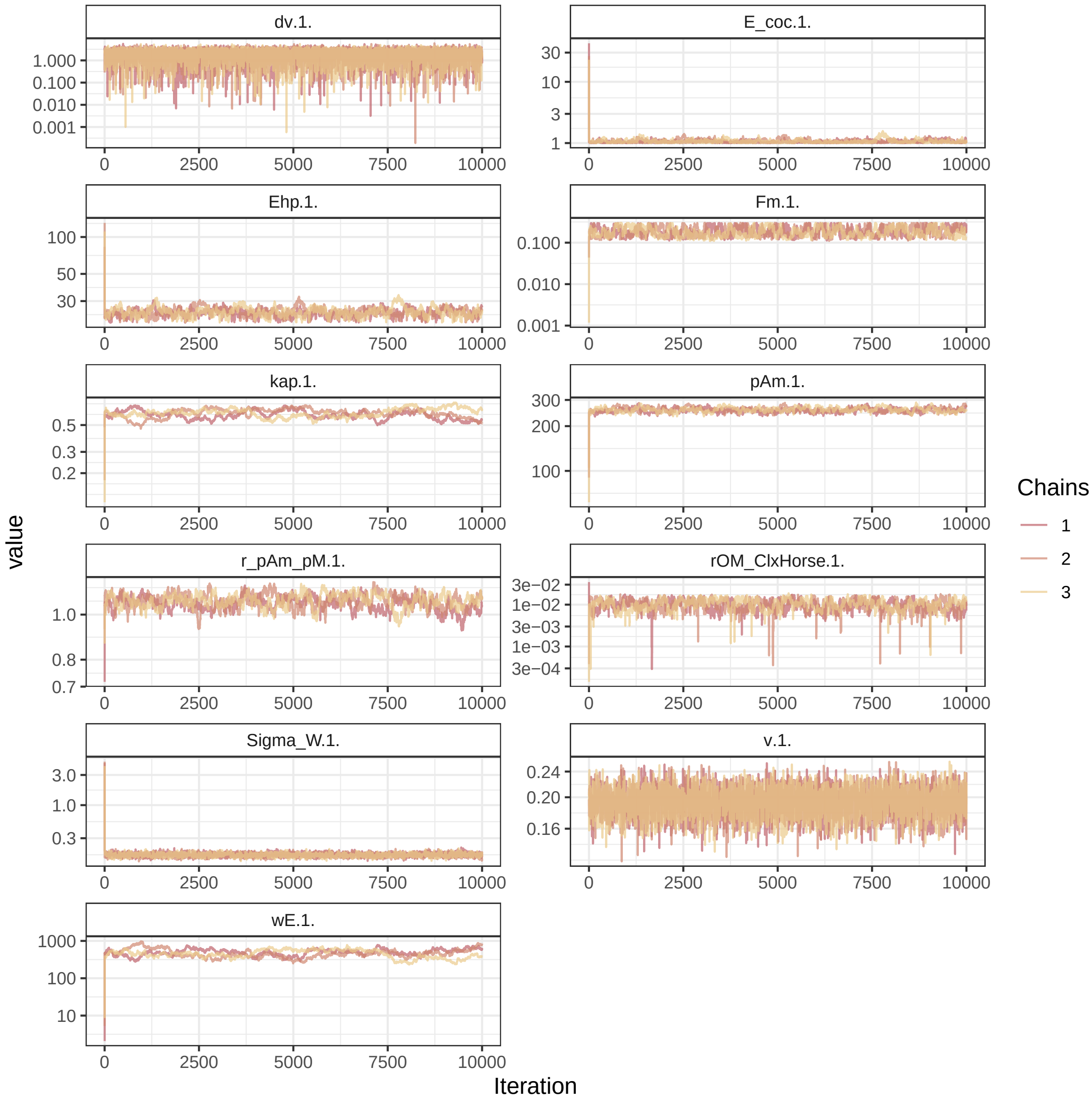


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

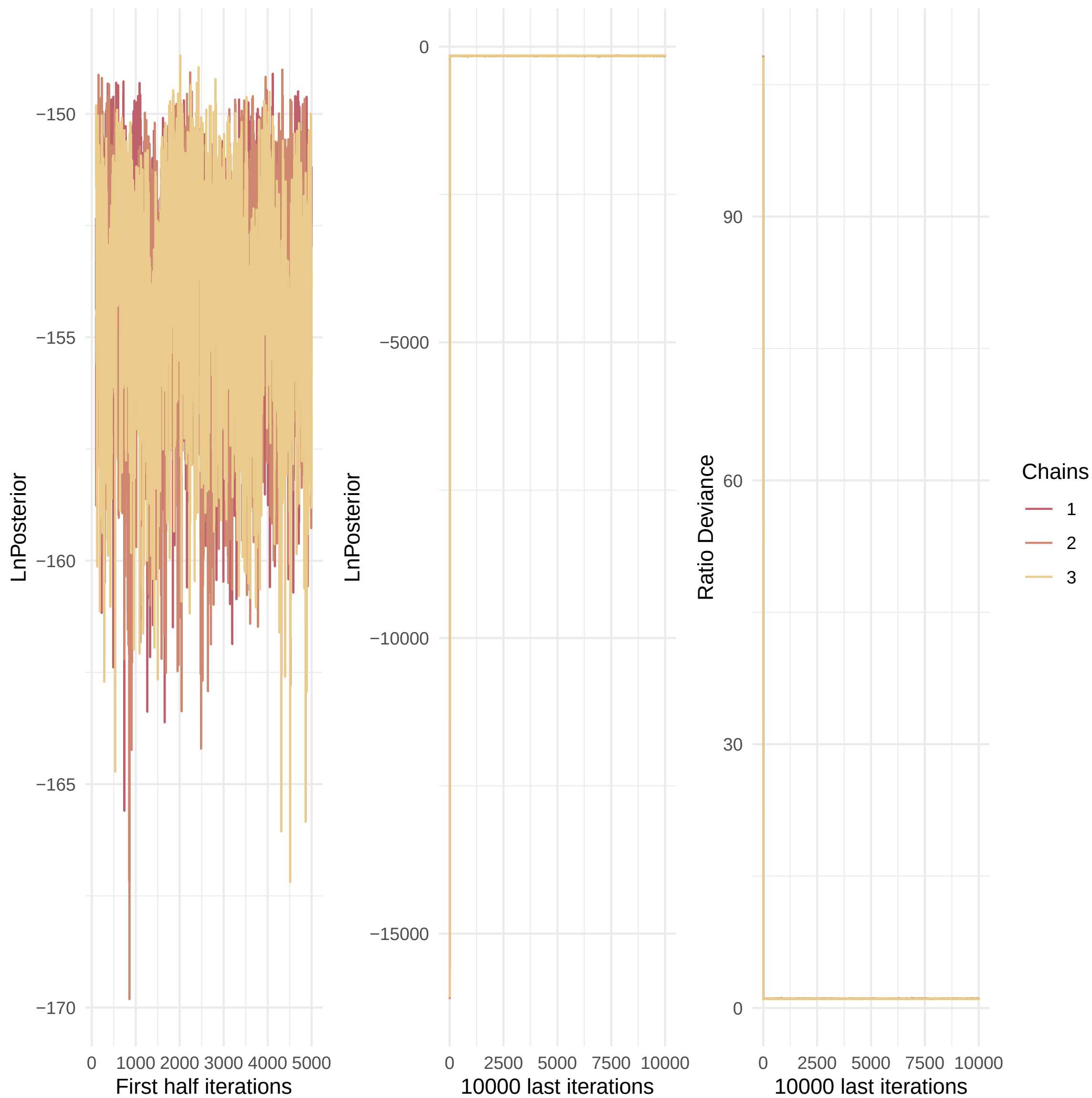


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

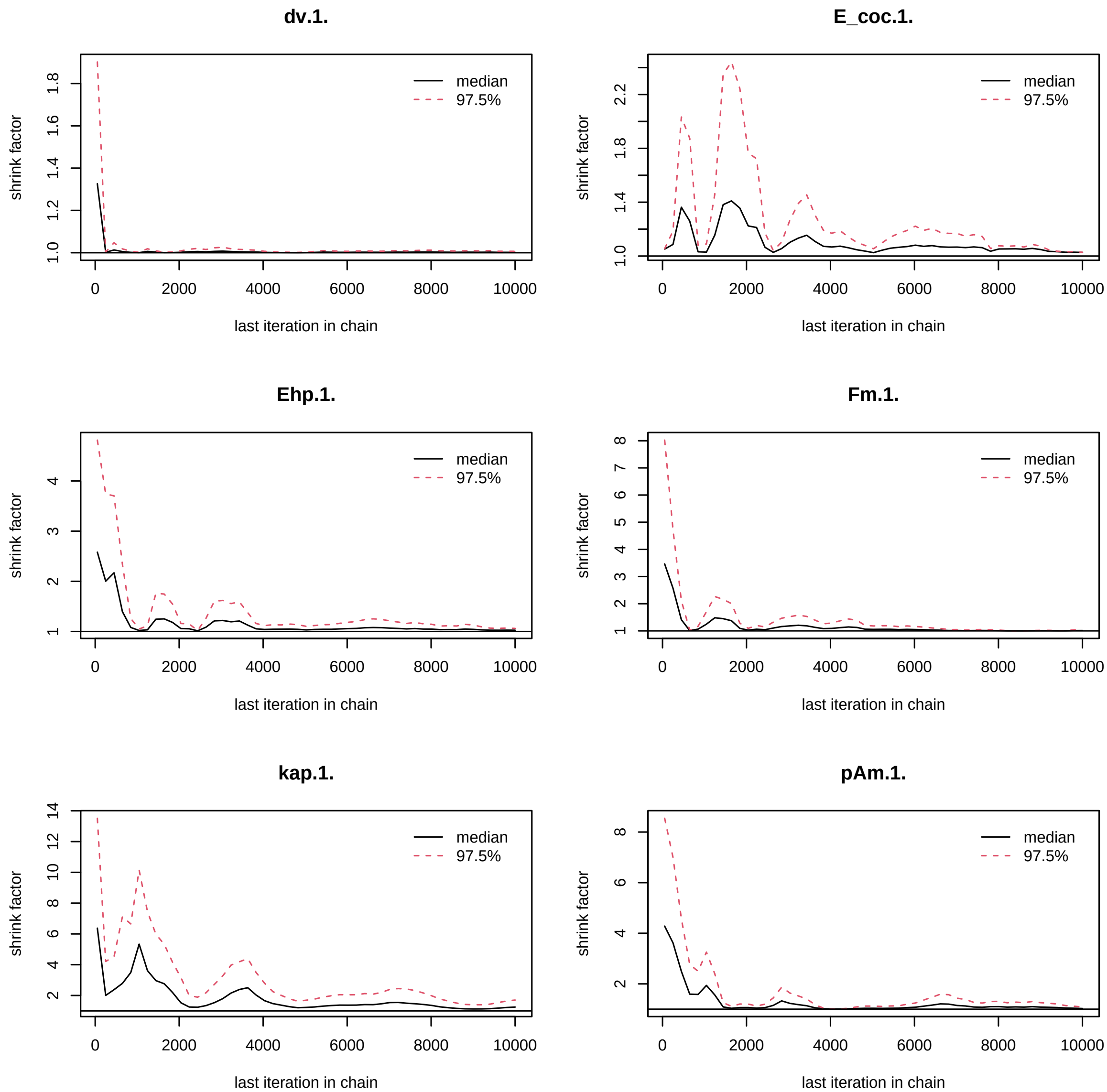


Figure 3: Chains convergence.

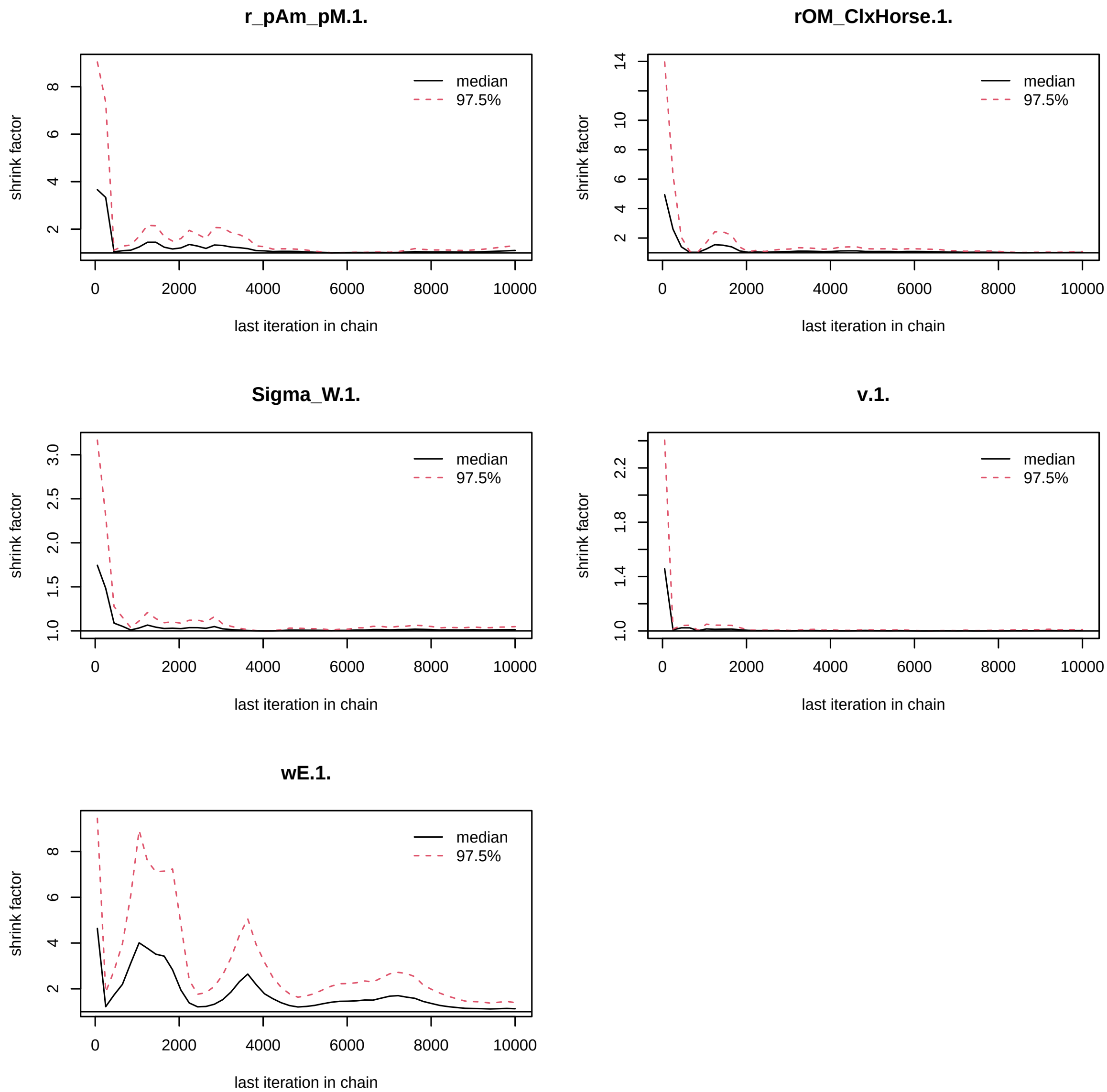


Figure 4: Chains convergence.

```r
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

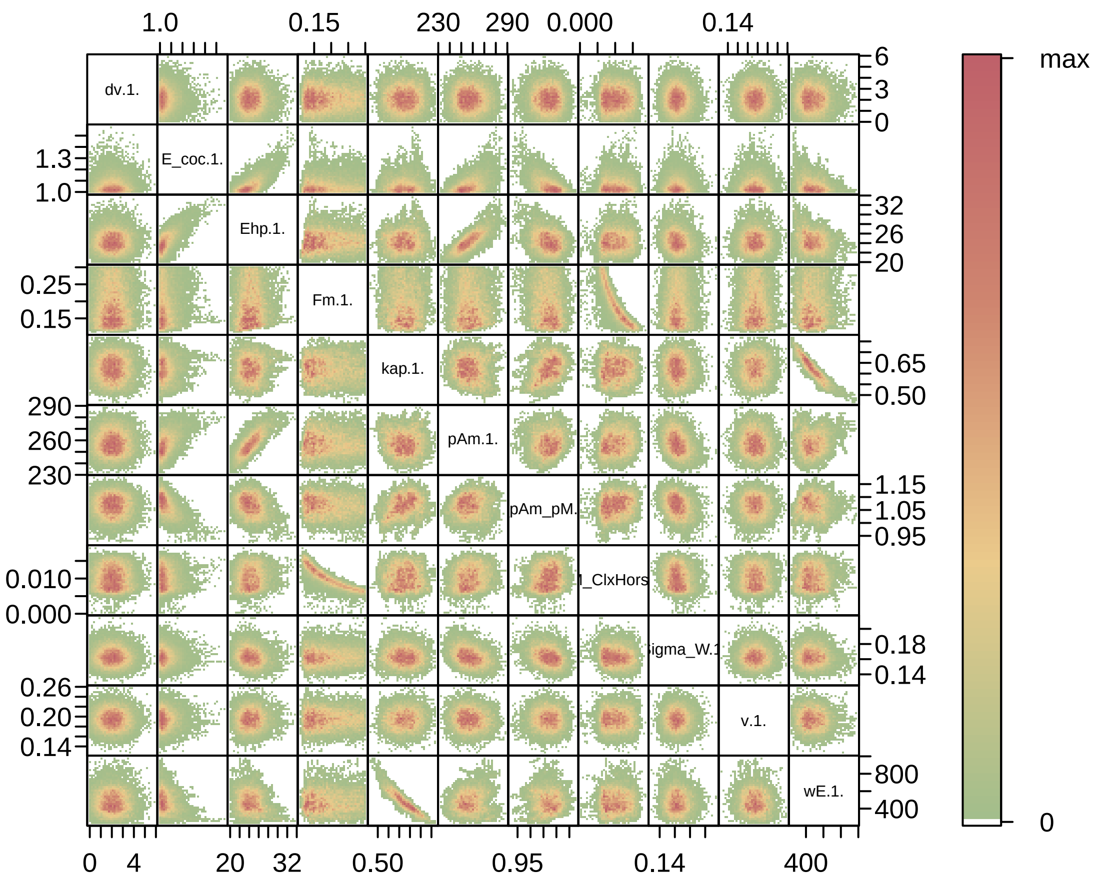


Figure 5: Parameter correlations.

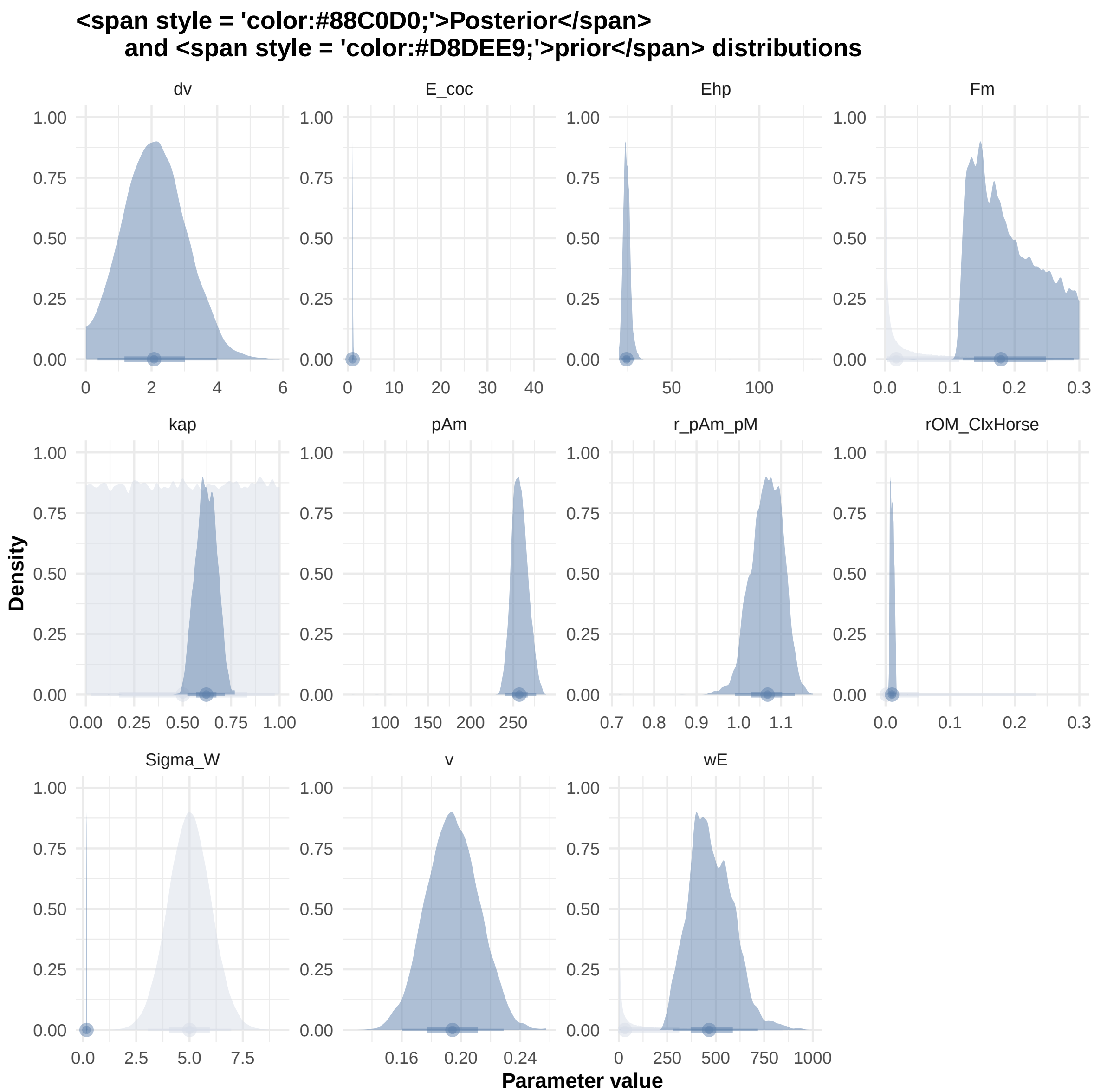


Figure 6: Distributions of the priors and the posteriors.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,
    kapreal,
```

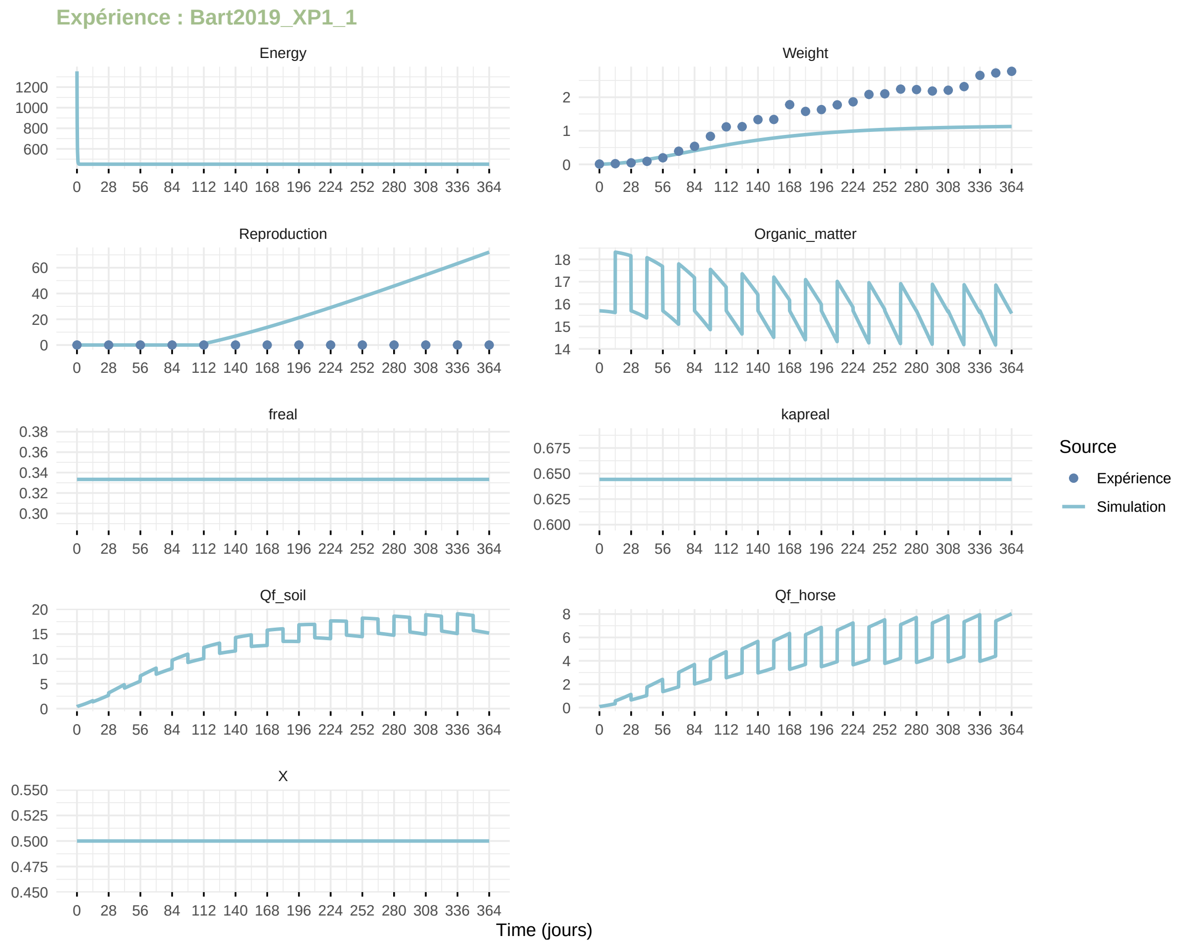
```
    freal,
    Qf_soil,
    Qf_horse,
    X"

step_sim <- 0.01

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pri

# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
```

```
    Reproduction <= 5e-6 ~ 0,
    .default = Reproduction
  )
)
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```



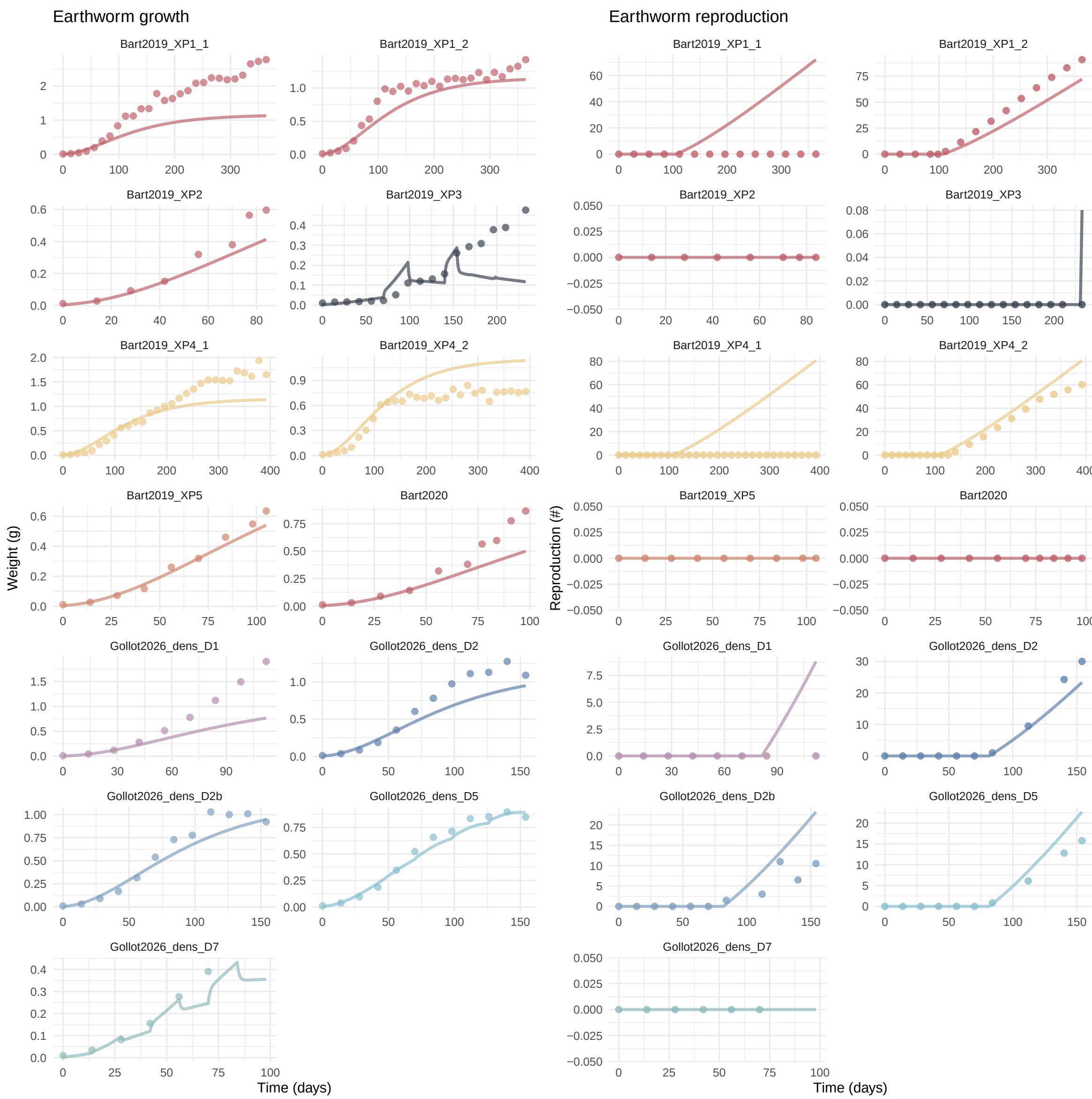
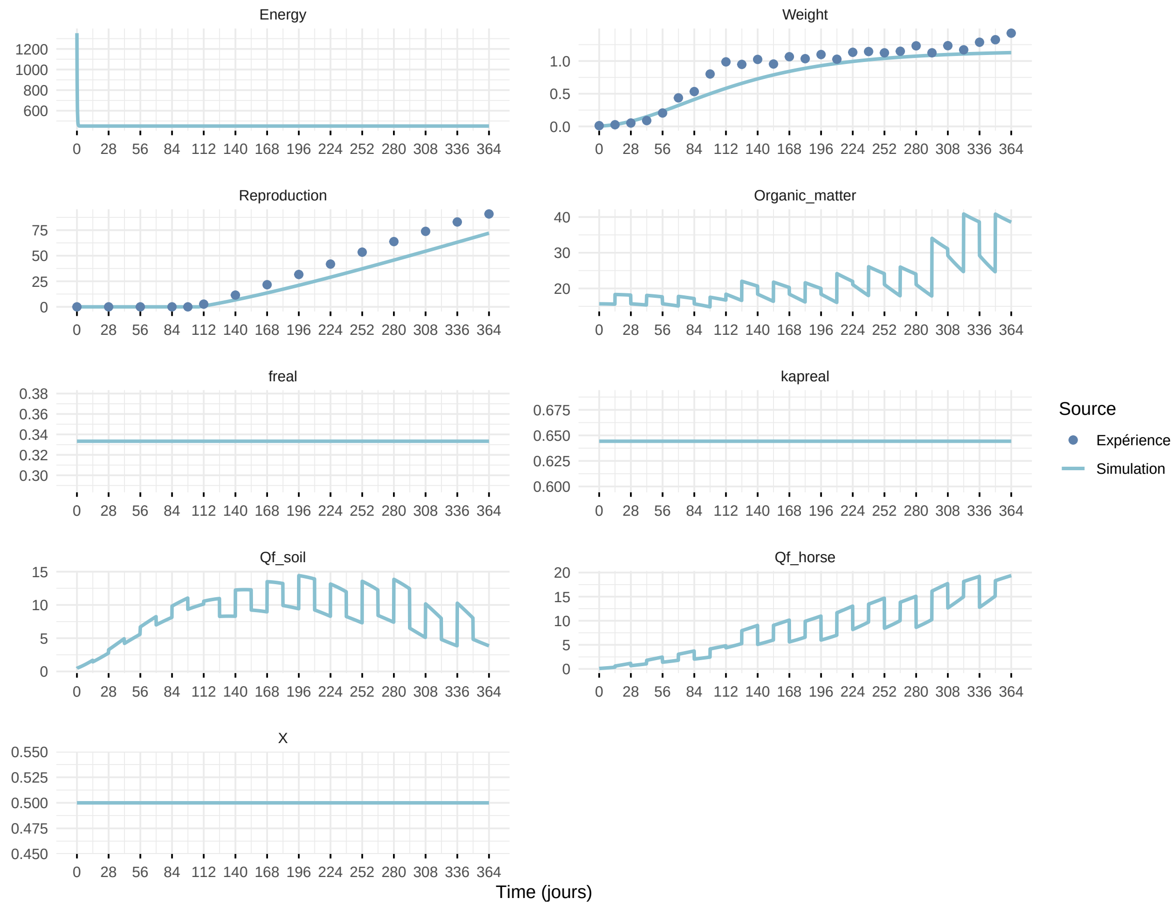
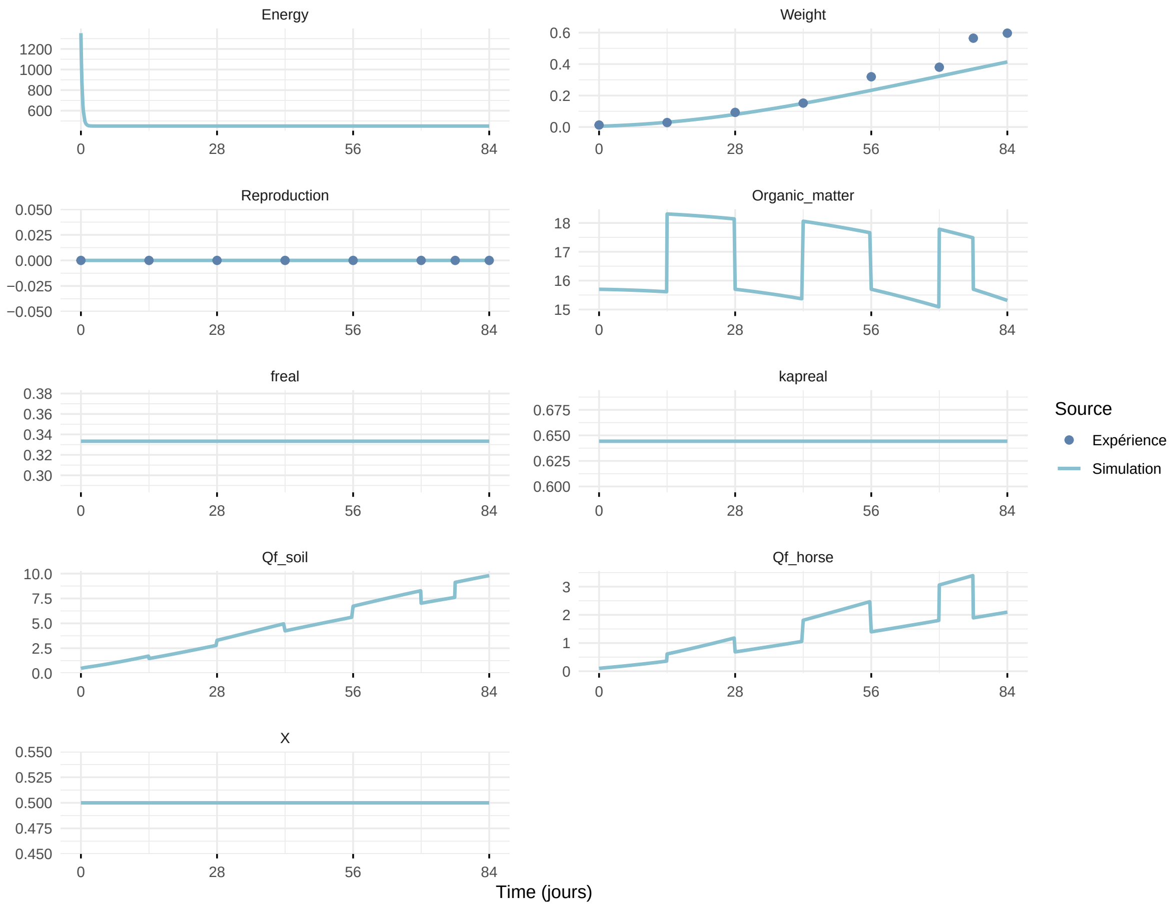


Figure 7: Simulations Weight and Reproduction.

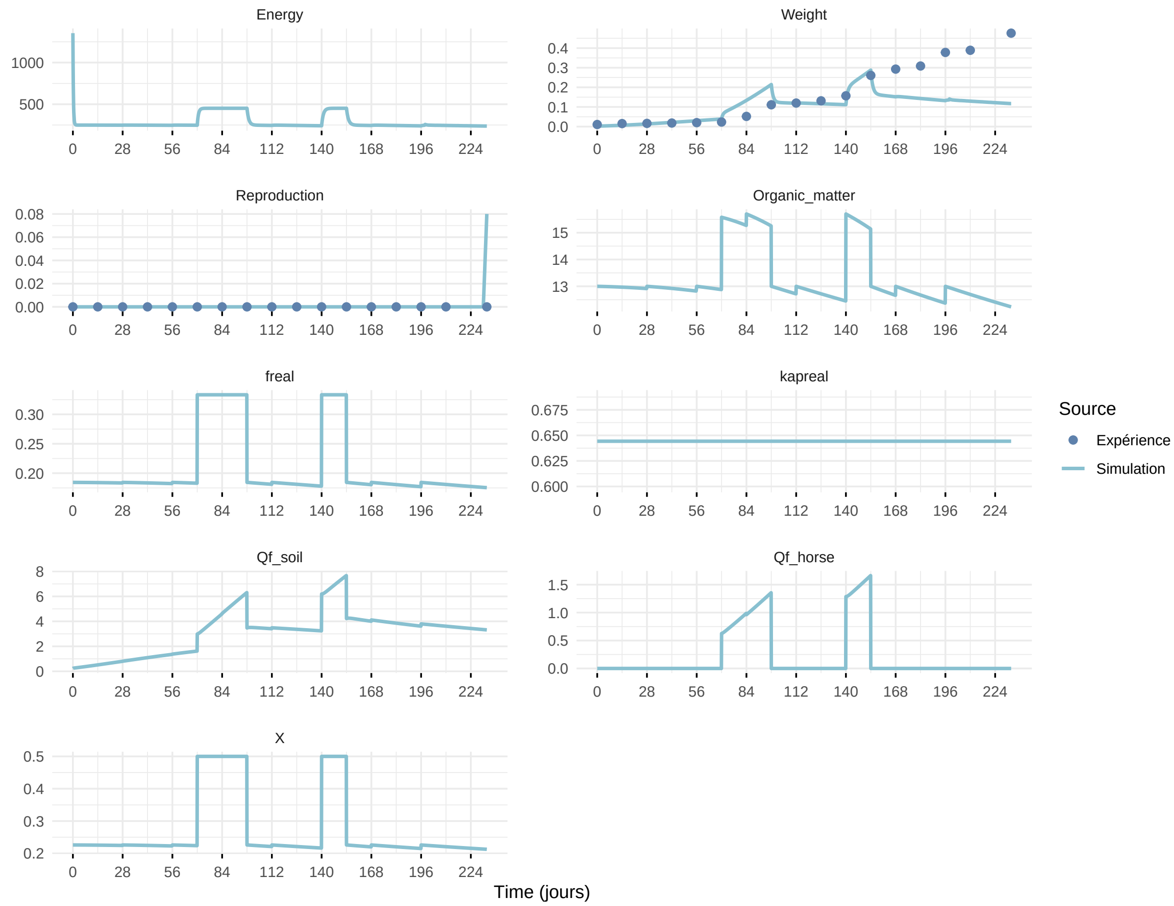
Expérience : Bart2019_XP1_2

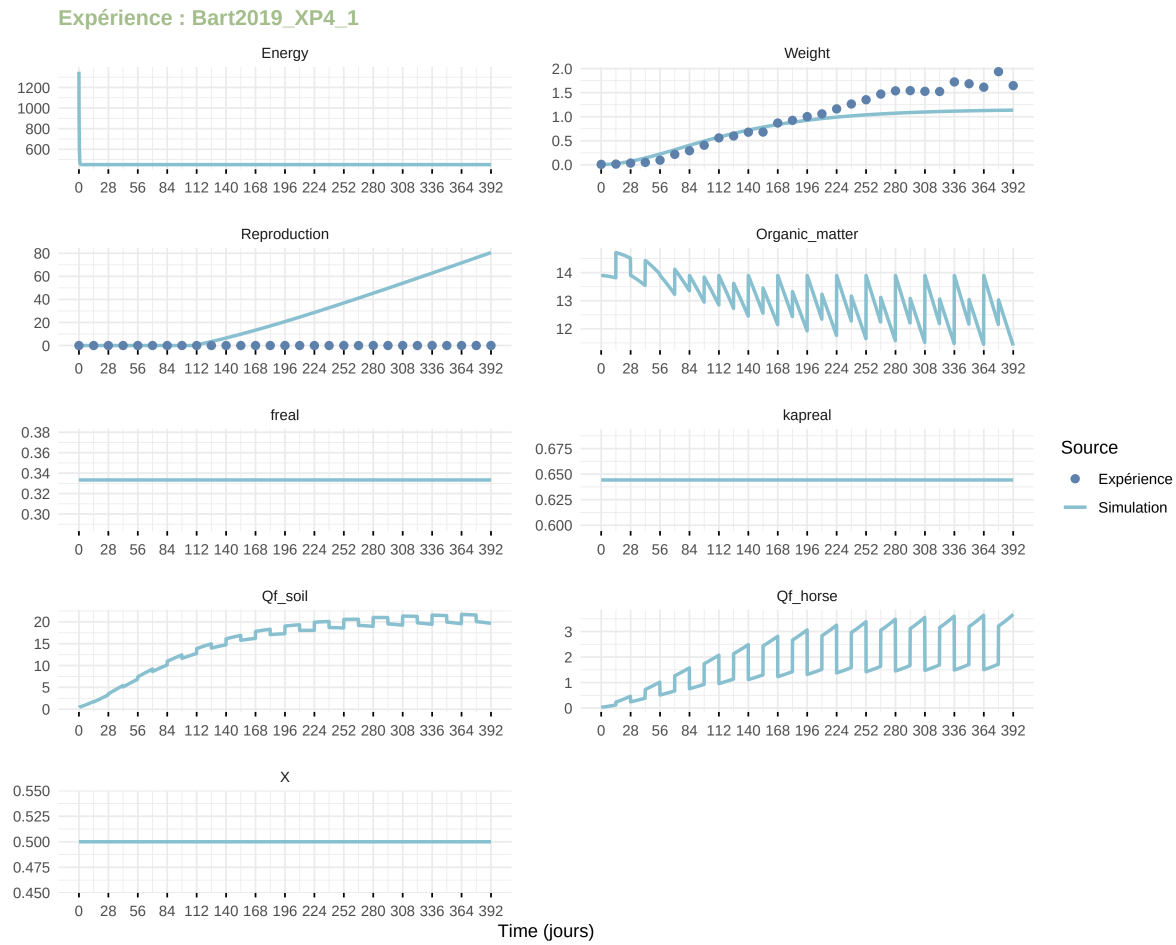


Expérience : Bart2019_XP2

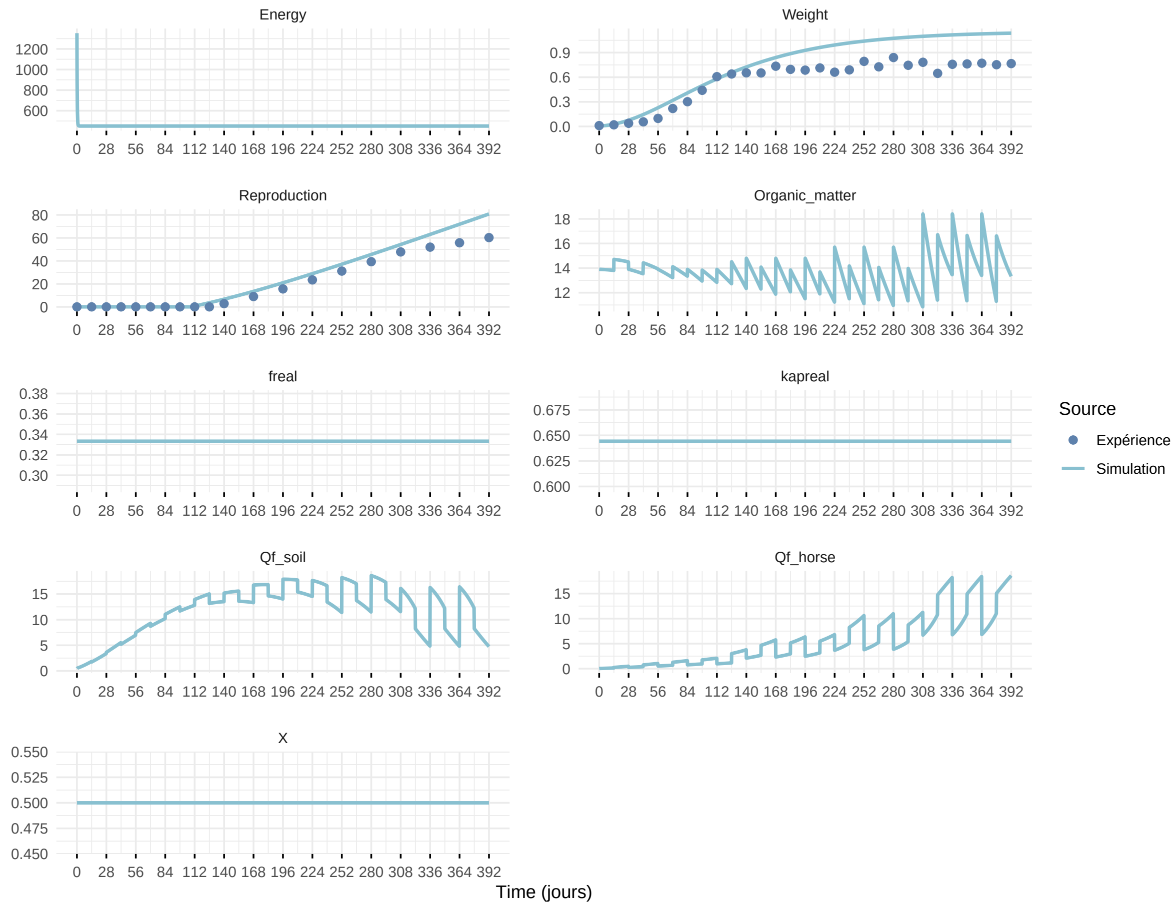


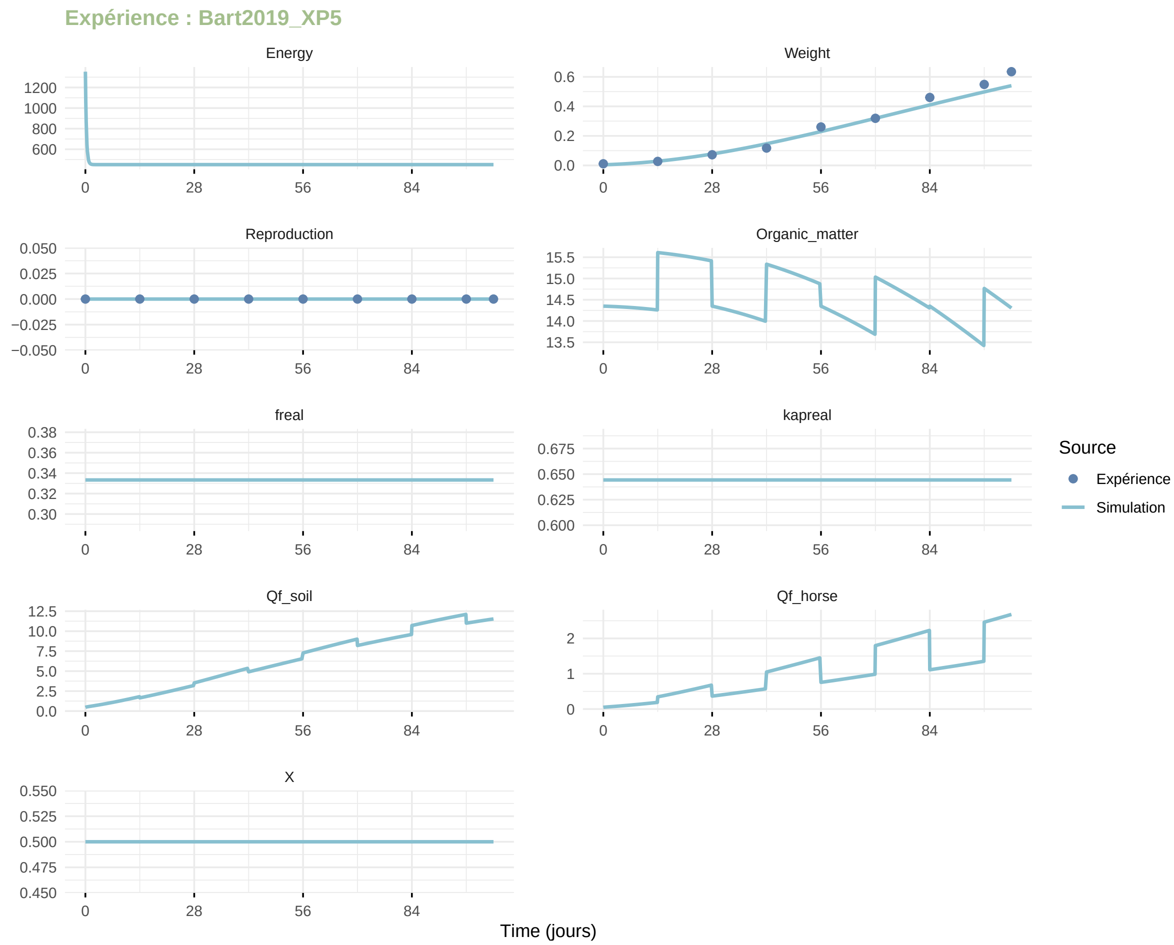
Expérience : Bart2019_XP3



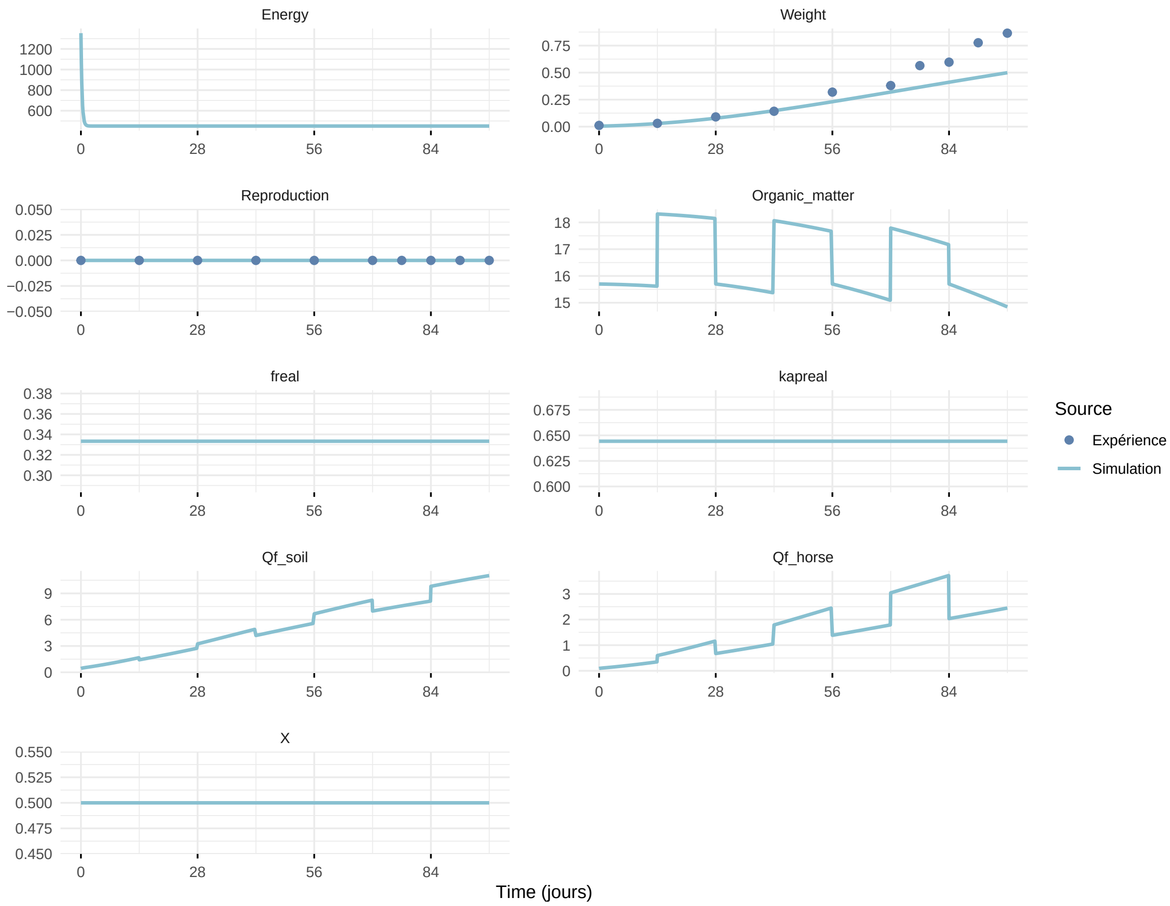

Expérience : Bart2019_XP4_1
Energy
1200
1000
800
600
Weight
2.0
1.5
1.0
0.5
0.0
Reproduction
80
60
40
20
0
Organic_matter
14
13
12
freal
0.38
0.36
0.34
0.32
0.30
kapreal
0.675
0.650
0.625
0.600
Qf_soil
20
15
10
5
0
Qf_horse
3
2
1
0
X
0.550
0.525
0.500
0.475
0.450
0 28 56 84 112 140 168 196 224 252 280 308 336 364 392
Time (jours)
Source
Expérience
Simulation

Expérience : Bart2019_XP4_2

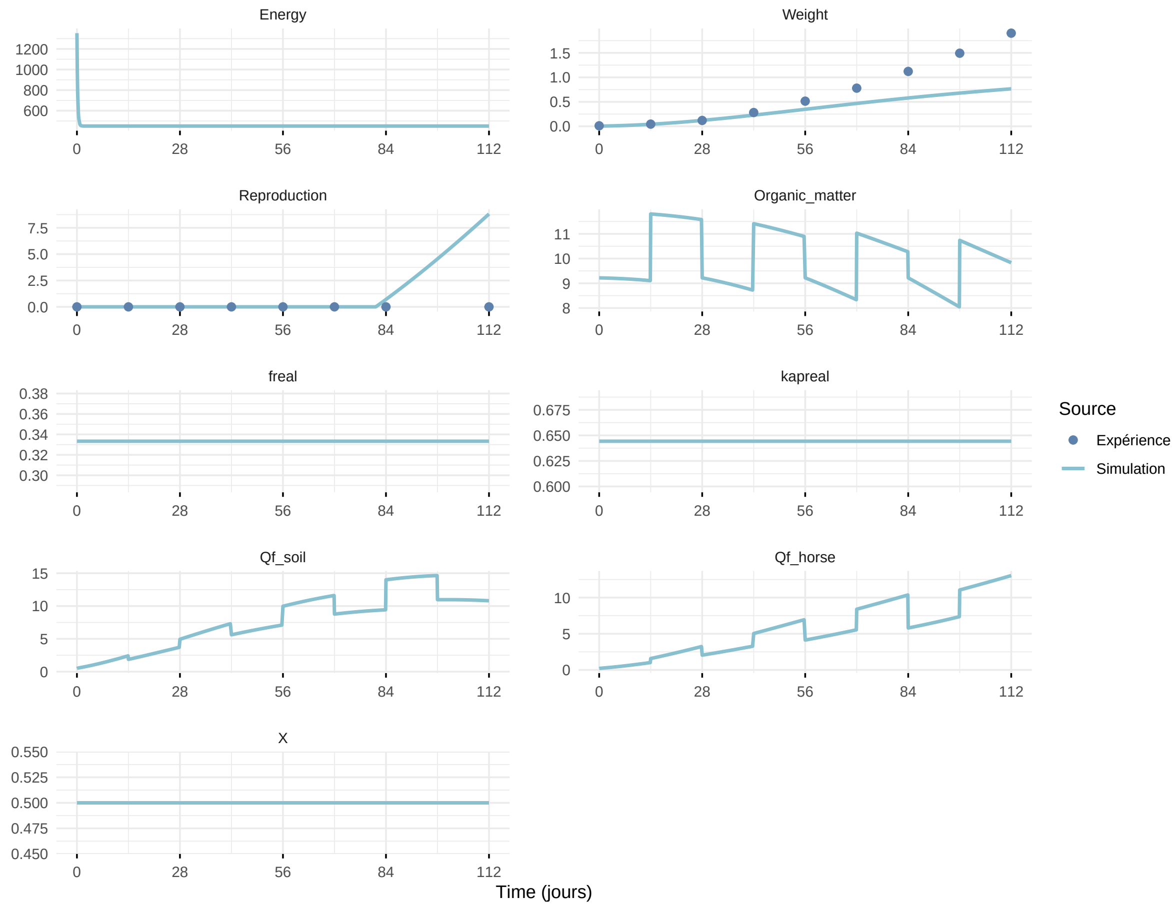



Expérience : Bart2019_XP5
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

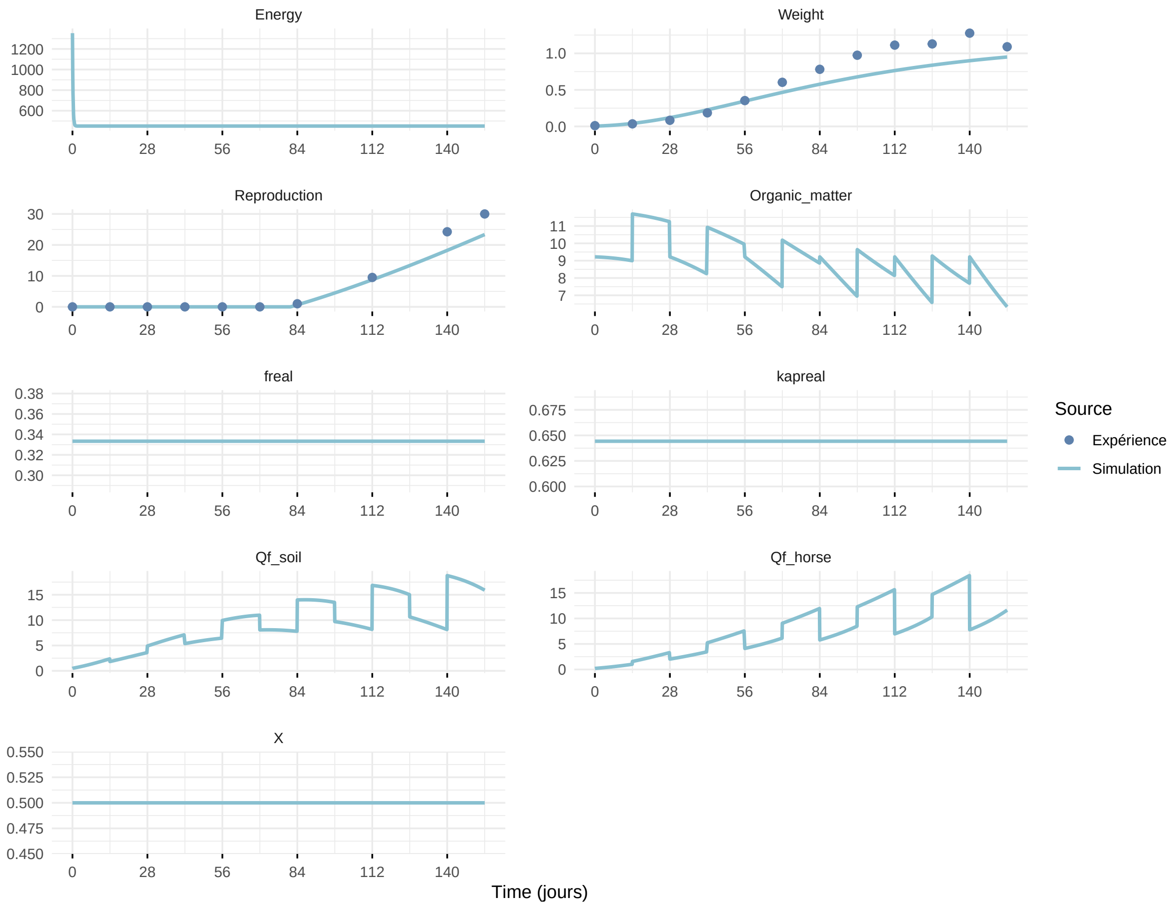
Expérience : Bart2020



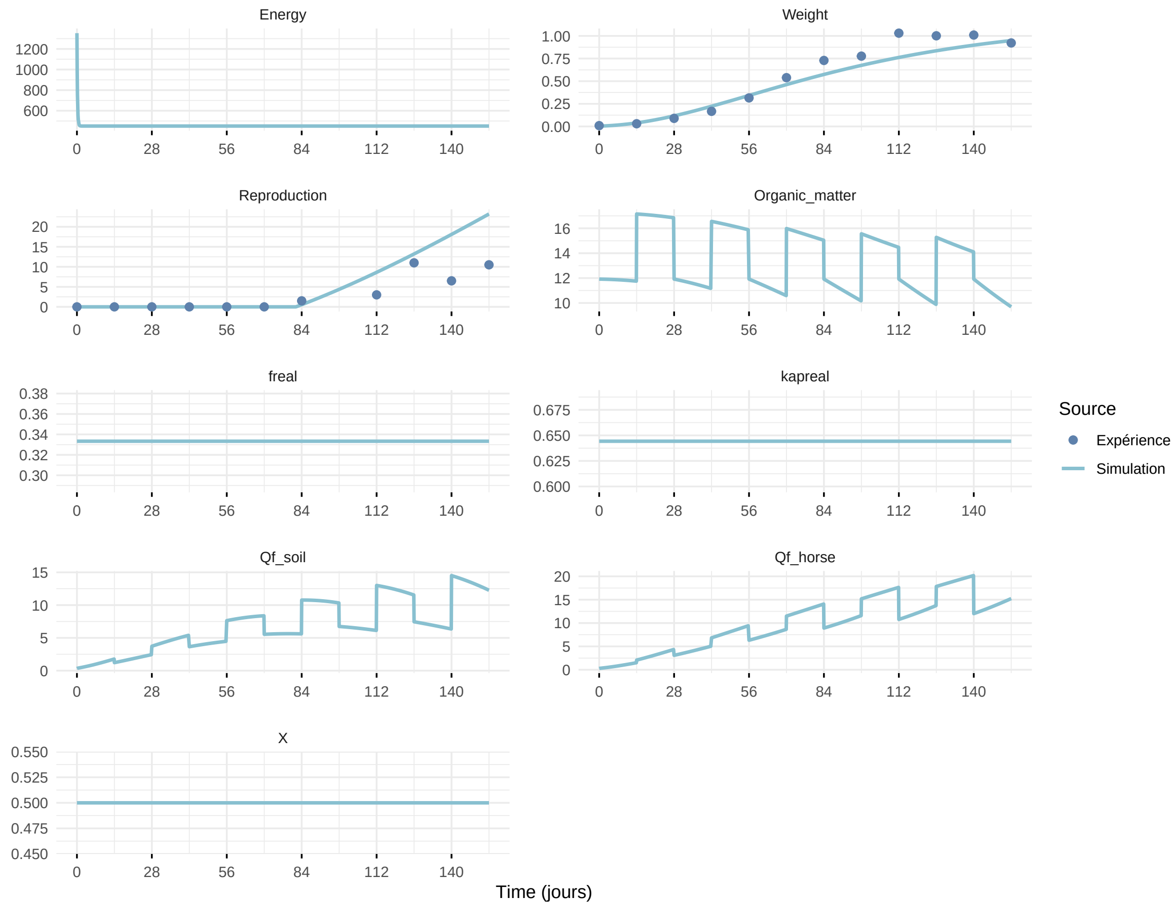
Expérience : Gollot2026_dens_D1

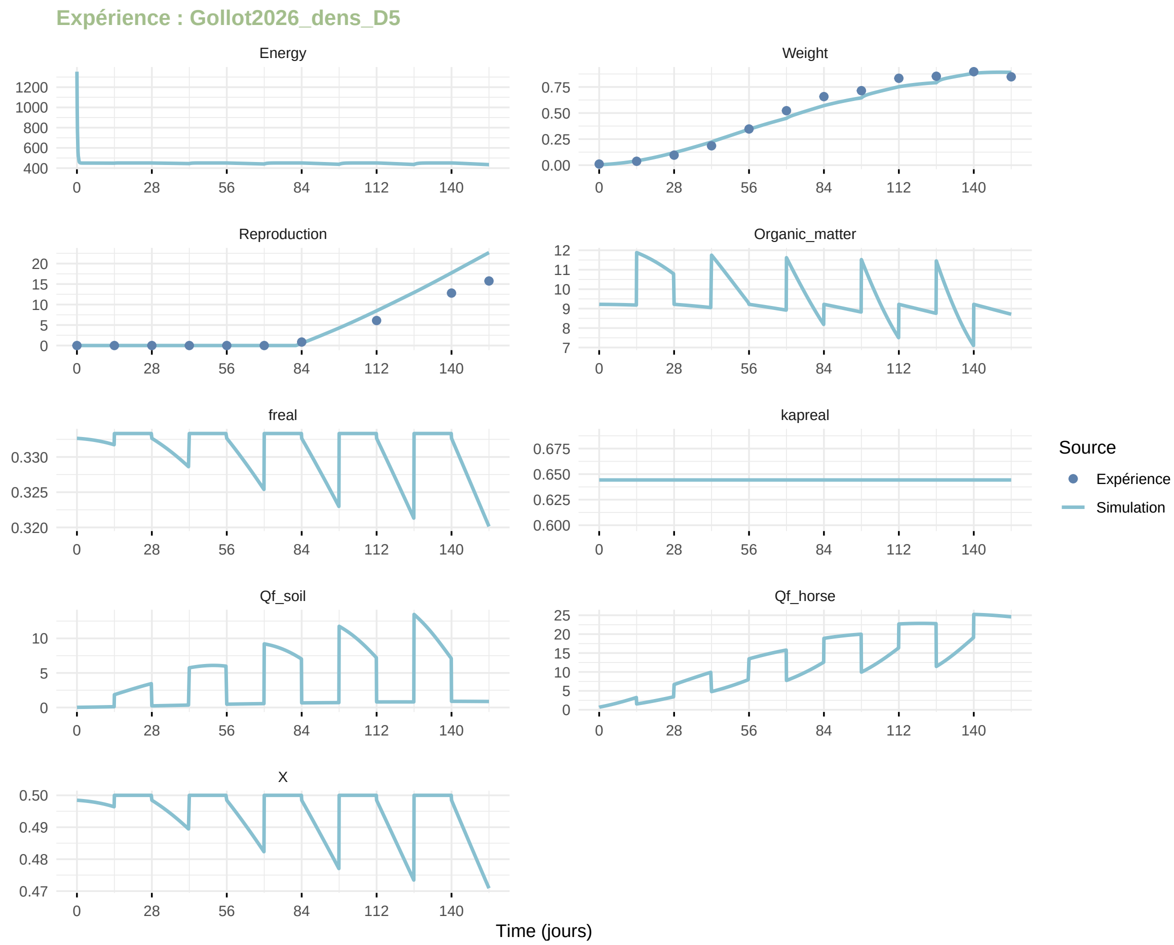


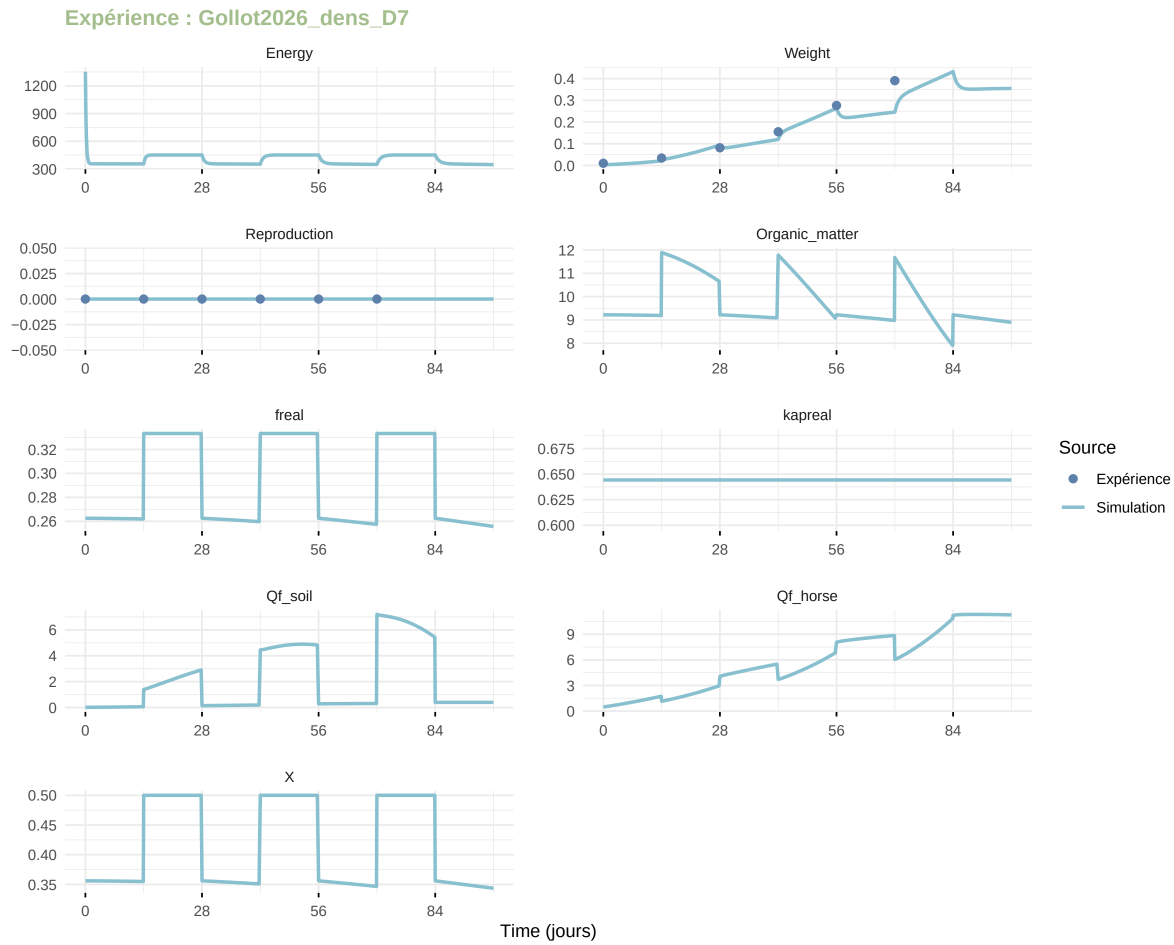
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b




Expérience : Gollot2026_dens_D5
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)


Expérience : Gollot2026_dens_D7
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)